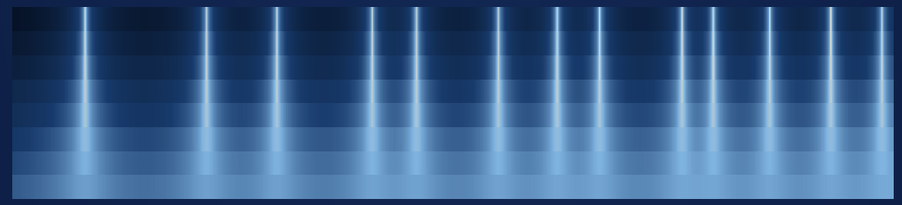
Running the Cauchy flow backwards

The Thorin measure of the reciprocal Xi function at any level is the Cauchy smoothing of the zeros, and at level one it is computable from the Euler product. The Riemann Hypothesis is the statement that this smoothing can be undone, down to the critical line, through positive measures. This volume computes that inverse problem as well as it can be done.

It evaluates the Xi function in ball arithmetic from Pólya's kernel alone, computes the de Bruijn Newman flow and its barrier method in Thorin form, and runs the backward Cauchy flow from the level one density, locating zeros and certifying window masses, by particle filters, nested sampling and the Monte Carlo criterion of Balazard, Saias and Yor.

It certifies Weil positivity in windows, follows truncations, friable, ferromagnetic and extreme value approximants and their zeros, and gathers the certificates that hold under the quasi Riemann hypothesis announced in October 2026. Each chapter says what a finite computation can decide and why no finite computation can decide the hypothesis itself. The code and data are at github.com/VadimSokolov/computing_rh.



the same measure at eight resolutions, from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top)

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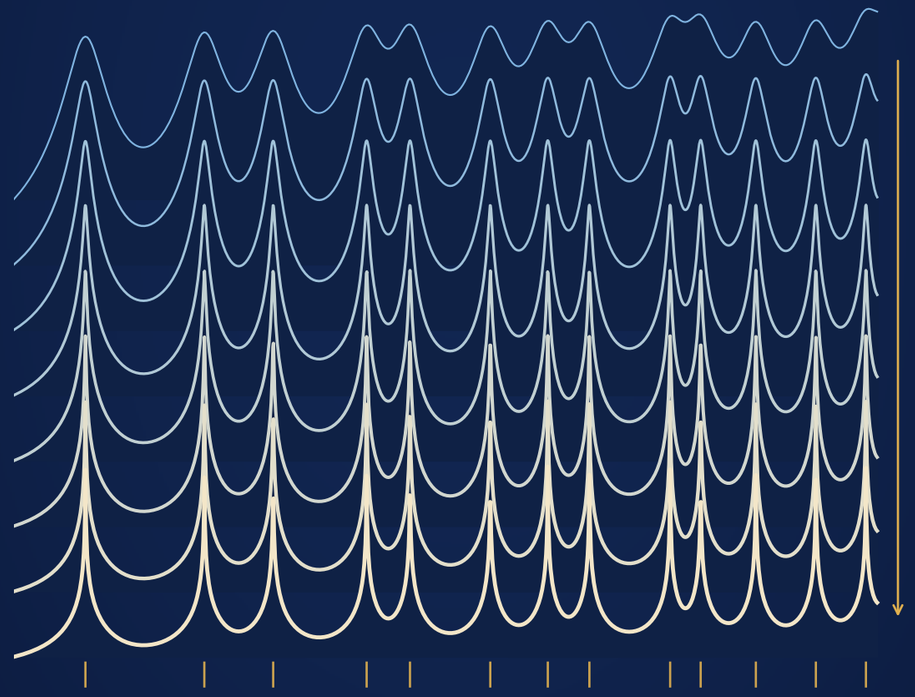
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Computing the Riemann Hypothesis

Volume II

Cauchy Deconvolution and Certified Computation



the backward Cauchy flow: from the level one density down to the zeros

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